TIC 257922637

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       4.476903 d  (ALIASED, see below)
    -> corrected period: 8.953807 d
  Epoch (t0)   3211.3193 BTJD
  Duration     0.1200 d
  Depth        6.932 %  (69,320 ppm)
  SNR          338.7

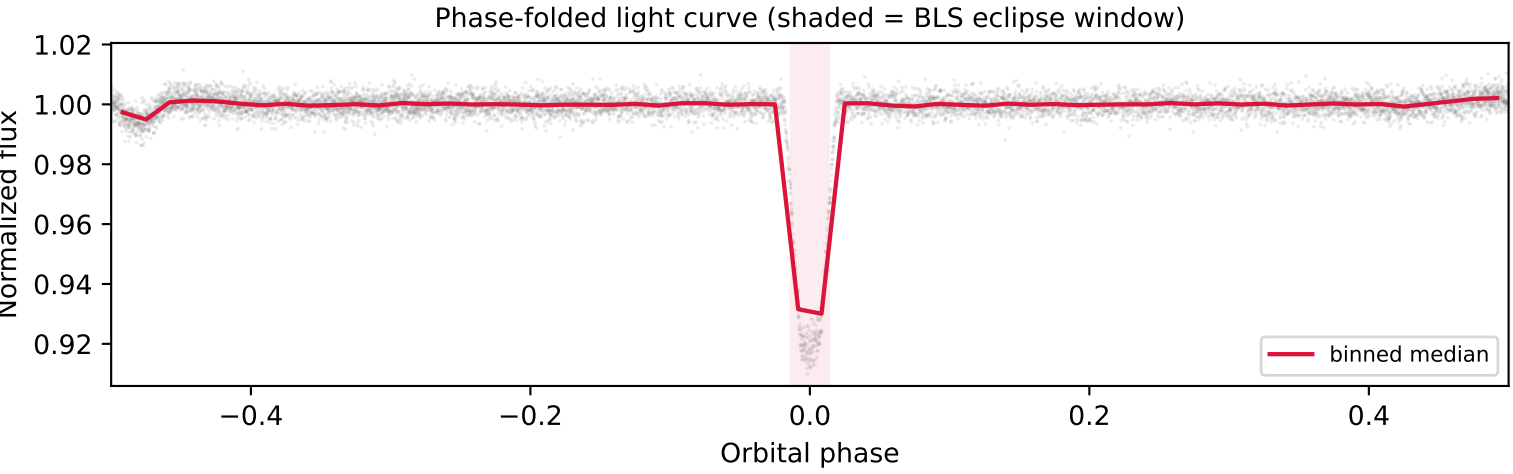
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     4  (gate >= 3)
  Robust depth ratio    1.09  (gate >= 0.50)
  Robust depth sigma    297.4  (gate >= 7.0)
  Odd/even z-score      3.72  (ALIAS FLAGGED)
  Secondary eclipse     detected, phase -0.47, 5.3 sigma
```

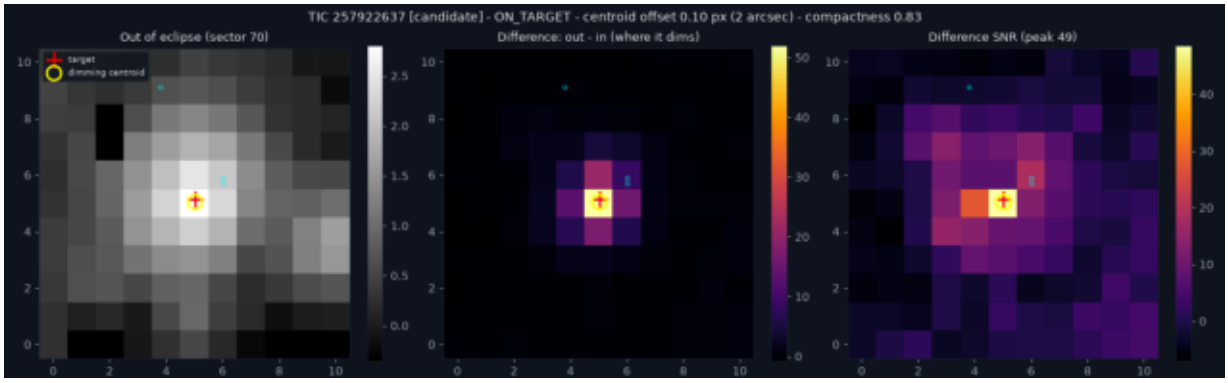
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 70    light-curve dip in this sector: 268.5 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   48.66   (gate >= 5.0)      Compactness   0.83   (gate >= 0.50)
  Dimming centroid offset     0.10 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: none within the cutout
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.72 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 8.953807 d, in any write-up.